



Student Name Answers

Eastern Goldfields College Mathematics Essential Unit 3 2018

Test 3

Working Time: 20 minutes

Marks: 19 marks

Calculator Free (No notes or calculator allowed)

Question 1 (4 marks – 1,1,1,1)

Circle the correct answer in each of the following multiple-choice questions.

a) The three sides of a right-angled triangle measure 40 m, 41 m and 9 m. The length of the hypotenuse is:

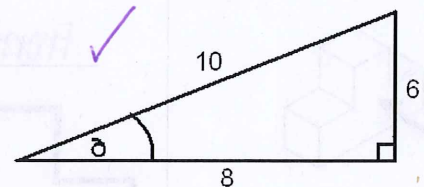
- (i) 9 m (ii) 40 m (iii) 41 m (iv) none of these

b) A map has a scale of 1 : 5000. 5cm on the map is equal to how far in real life?

- (i) 5000 cm (ii) 2500 cm (iii) 25 m (iv) 250 m

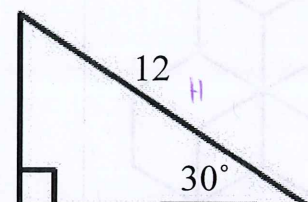
c) In relation to the angle, which is the opposite side?

- (i) 10 (ii) 8 (iii) 6 (iv) none of these



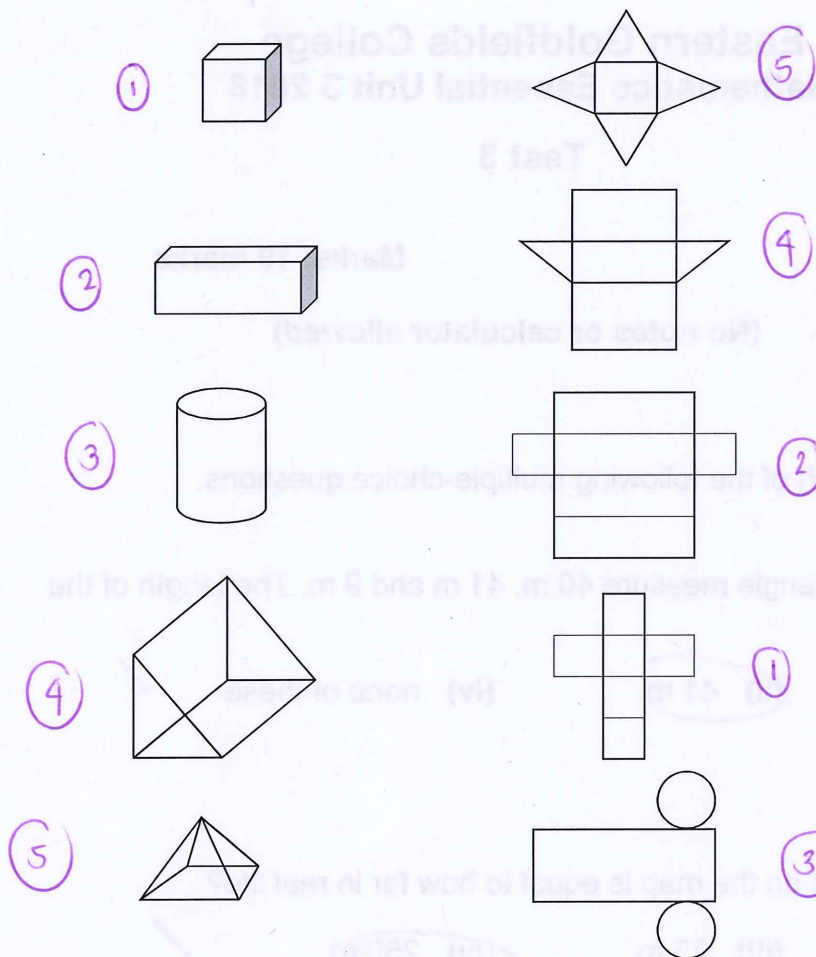
d) The value for x in the diagram is given by:

- (i) $12 \times \sin 30^\circ$ (ii) $12 \times \cos 30^\circ$ (iii) $\frac{12}{\tan 30^\circ}$ (iv) $\frac{12}{\sin 30^\circ}$



Question 2 (2 marks)

Match each shape to its net

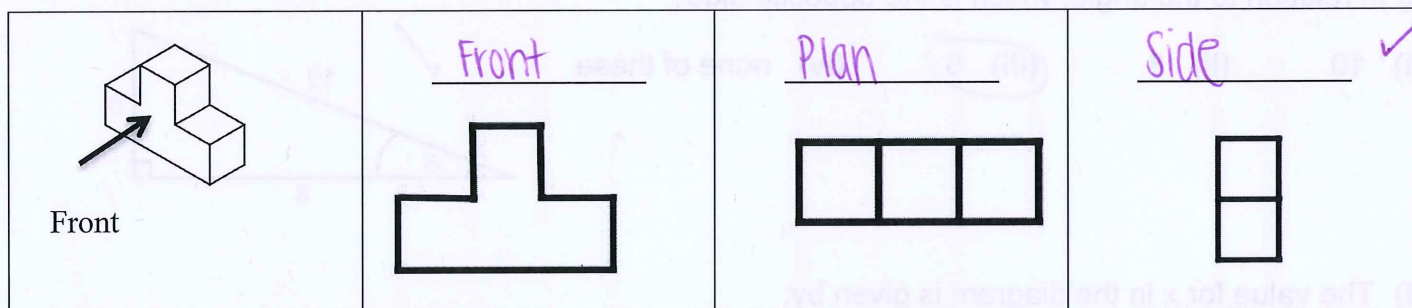


✓✓ All correct

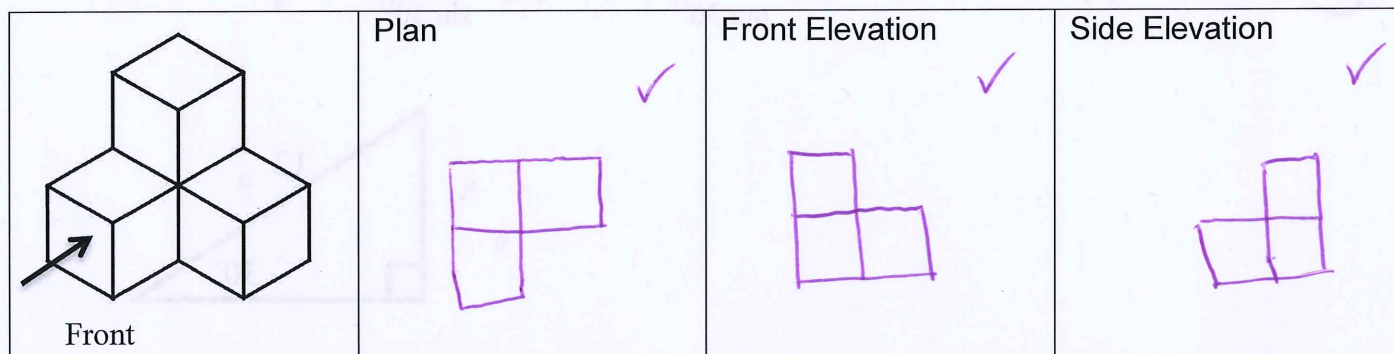
✓ ~2 wrong.

Question 3 (4 marks – 1,3)

a) Label the different views for the following 3D shape.



b) Draw the shape in its different views.



✓ 6

Question 4 (3 marks)

Miss Jones, Mr Elliot and Mr Cook have all constructed pillow forts as seen in the diagram below. They wish to run cable to all 3 forts so that they can connect their laptops. How much total cable will be needed?



DON'T WORRY.

8m



6m



$$x = \sqrt{8^2 + 6^2} \quad \checkmark$$

$$x = 10$$

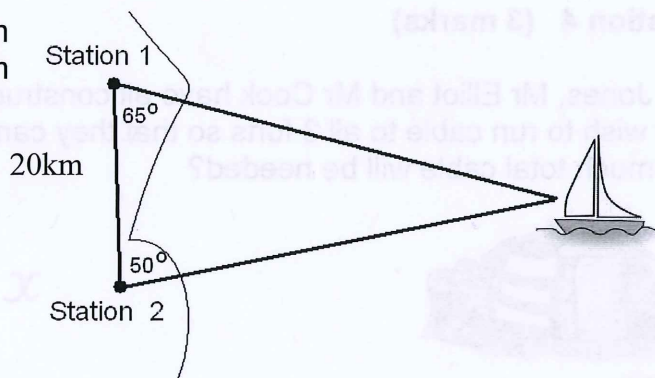
$$10 + 8 + 6 \quad \checkmark$$

$$= 24m \quad \checkmark$$

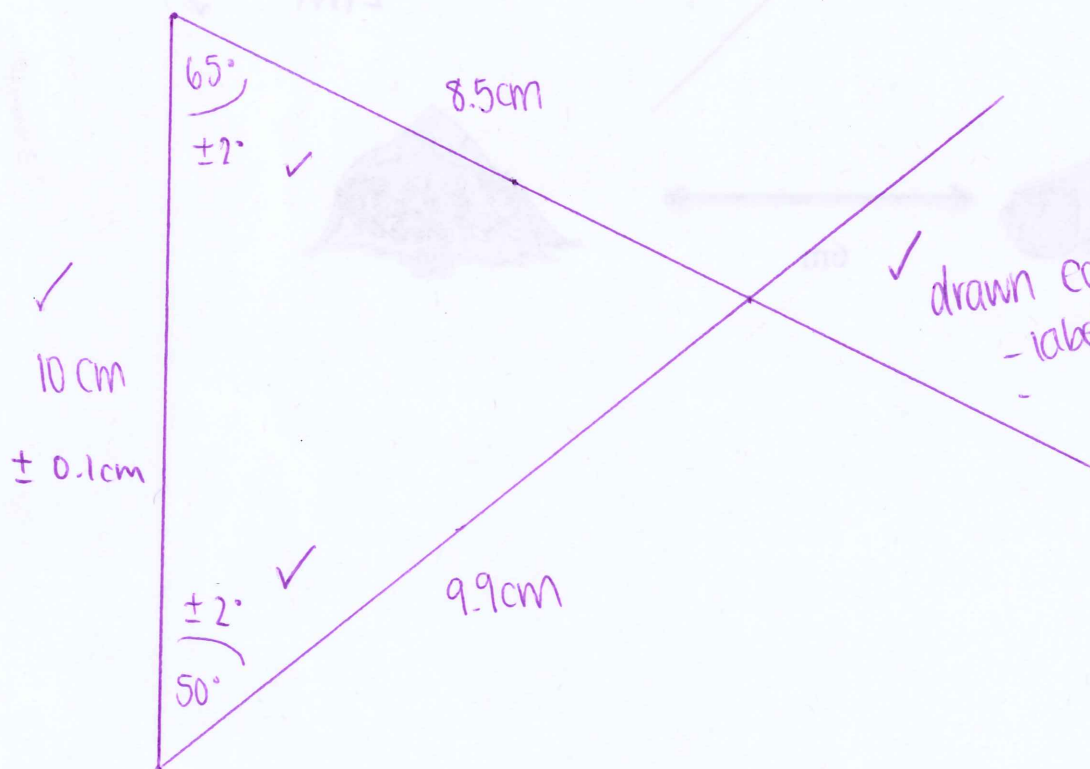
Question 5 (6 marks)

Create an accurate scale drawing of the diagram on the right to determine how far the boat is from each station.

Use scale 1 = 200000



- 1 lines dont intersect
- not labelled -



$$8.5 \times 2 = 17 \text{ km}$$

$$9.9 \times 2 = 19.8 \text{ km}$$



Student Name _____

Eastern Goldfields College Mathematics Essential Unit 3 2018

Test 3

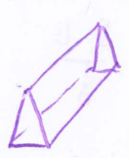
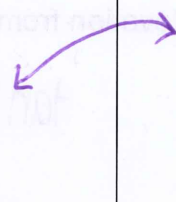
Working Time: 35 minutes

Marks: 29 marks

Calculator Assumed (Formulae sheet and one A4 page of notes)

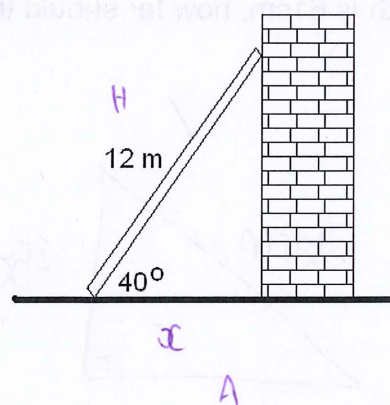
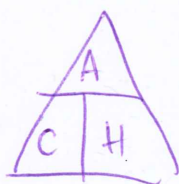
Question 6 (2 marks)

Draw and name the following shape, according to the description given below:

Description	Name	Drawing
A 3D shape that is made up of 5 faces – 2 triangles and 3 rectangles.		 triangular prism

Question 7 (3 marks)

A ladder 12 m long leans against a wall. If it forms an angle of 40° with the ground, how far is the bottom of the ladder away from the wall?



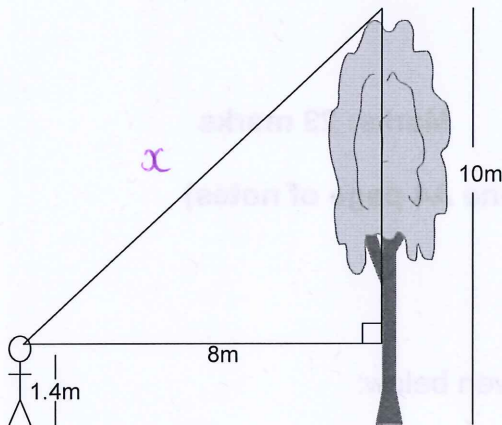
$$\cos 40 = \frac{x}{12}$$

$$\cos 40 \times 12 = x$$

$$9.2 \text{ m} = x$$

Question 8 (5 marks – 3,2)

- a) A boy notices a bird sitting at the very top of a 10m tall tree. If he is standing 8m from the base of the tree, what is the distance between his eye and the top of the tree?



$$x = \sqrt{(8^2 + 10^2)} \quad \checkmark$$

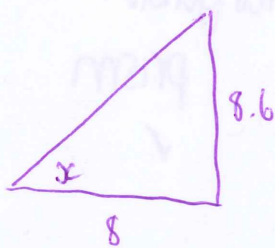
$$x = 11.75 \text{ m} \quad \checkmark$$

OR

$$\sqrt{(8^2 + 10^2)} \quad \checkmark$$

$$= 12.8$$

- b) What is the angle of elevation from the boy's eye to the top of the tree?

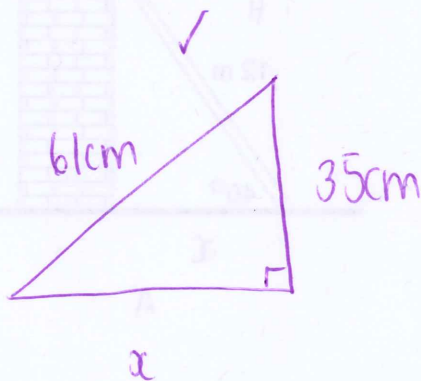


$$\tan \theta = \frac{8.6}{8} \quad \checkmark$$

$$\theta = 47^\circ \quad \checkmark$$

Question 9 (3 marks)

The school council needs to have a ramp build over the steps of each of the building exits, to accommodate a student in a wheelchair. If the school building is 35cm off the ground and the ramp length is 61cm, how far should the steps reach out to accommodate the ramp?

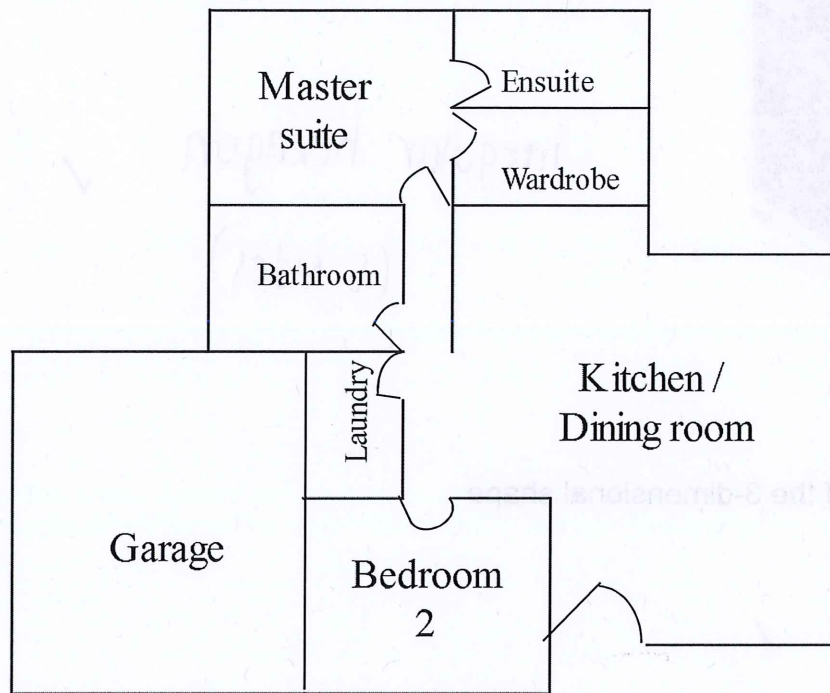


$$\sqrt{(61^2 - 35^2)} \quad \checkmark$$

$$x = 50 \text{ cm} \quad \checkmark$$

Question 10 (7 marks: 2, 2, 3)

Below is the floor plan for Mrs Wood's new holiday house.



Scale 1: 200

- a) Using the scale given, find the area of the garage

$$4.5 \times 3.9$$

$$900 \times 780 = 702000 \text{ cm}^2 \quad \text{or} \quad 70.2 \text{ m}^2 \quad \checkmark \checkmark$$

- b) Mrs Wood decides that she wants to concrete the floor of the garage. If concrete costs \$21.20 per square metre, how much will it cost to complete?

$$70.2 \times \$21.20 = \$1488.24 \quad \checkmark$$

↑
FT

- c) The bathroom, laundry and ensuite are to be tiled, with tiles costing \$41.40/m². How much will it cost to buy tiles?

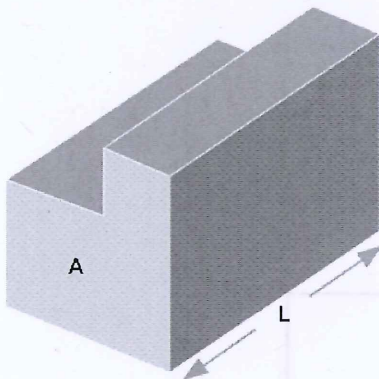
$$\begin{aligned} \text{laundry: } 2 \times 1.3 & \quad 400 \times 260 & = 447200 \text{ cm}^2 & \checkmark \\ \text{bathroom: } 2 \times 2.6 & \quad 400 \times 520 & & \\ \text{Ensuite: } 2.6 \times 1.3 & \quad 520 \times 260 & & \end{aligned}$$

$$44.72 \text{ m}^2 \times 41.40 = \$1851.42 \quad \checkmark$$

* accept 40-45 m² = \$1656 → \$1863

Question 11 (4 marks – 1, 3)

a) What is the name of the front elevation (face A)



irregular hexagon

(6 sides)

✓

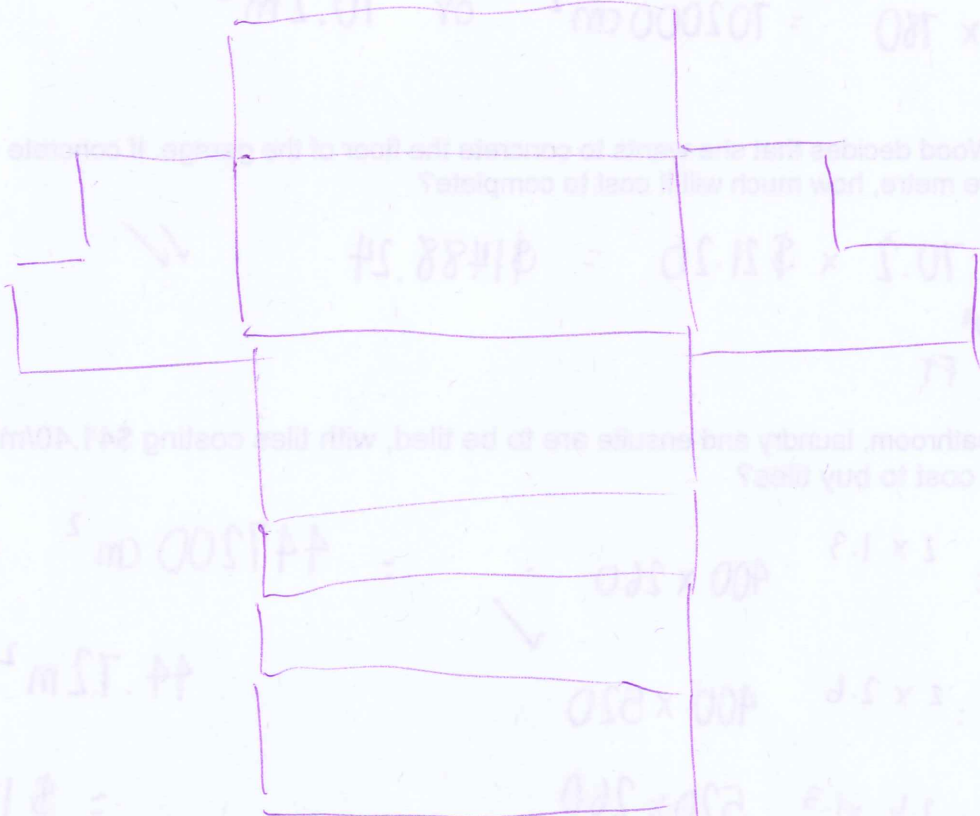
b) Draw the net of the 3-dimensional shape

8 faces

✓

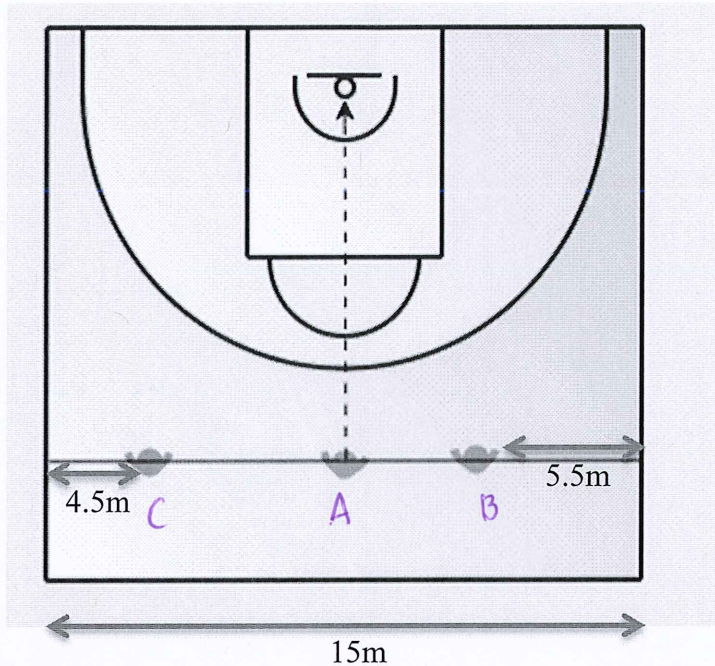
drawn correctly

✓



Question 12 (5 marks)

A basketball coach draws a line across the court that is exactly 7m away from the hoop (as seen in the diagram below). He gets 3 players to stand on the line for a shooting drill; one is directly in front of the hoop with the other 2 players set up either side. One of the players protests and says that this setup is not fair. Is the player correct? Justify with calculations



A : 7m away ✓

B : $\sqrt{7^2 + 2^2}$ ✓
 $= 7.3\text{m}$

C : $\sqrt{7^2 + 3^2}$ ✓
 $= 7.6\text{m}$

✓
yes, players are different
distances from goal ✓

END OF TEST

5

